Benchmarking Computational Doublet-Detection Methods for scRNA-seq Data Project Proposal

1. Group Information

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2. Problem Statement

In droplet-based single-cell RNA-seq, a doublet forms when two or more cells are captured in one droplet and appear to be a single cell. To enable further work on the data, doublets need to be removed, so computational detection is widely used. The most comprehensive benchmark of detection methods dates from 2021 (Xi and Li) and found no method consistently best across datasets. Several new tools have been developed since, such as Chord and vaeda. A 2024 (Zhang et al) benchmark evaluated only a few tools and limited metrics, and ranked tools differently on barcoded data, partly depending on how test doublets were simulated. A neutral, comprehensive comparison of current tools would help users choose a method for their data type.

3. Tools and Justification

Tool	Reference	Link	Justification
Scrublet	Wolock, S. L., et al. "Scrublet: Computational Identification of Cell Doublets in Single-Cell Transcriptomic Data." *Cell Systems* 8.4 (2019): 281–291.	github.com/AllonKleinLab/scrublet	Python kNN artificial-doublet method
DoubletFinder	McGinnis, C. S., et al. "DoubletFinder: Doublet Detection in Single-Cell RNA Sequencing Data Using Artificial Nearest Neighbors." *Cell Systems* 8.4 (2019): 329–337.	github.com/chris-mcginnis-ucsf/DoubletFinder	Highest mean AUPRC and usability score in Xi and Li; Seurat/R
scDblFinder	Germain, P.-L., et al. "Doublet Identification in Single-Cell Sequencing Data Using scDblFinder." *F1000Research* 10 (2022): 979.	github.com/plger/scDblFinder	Best overall in Xi and Li's protocol extension but lowest AUPRC on barcoded data in Zhang et al.; tuned on the 16 benchmark datasets
scds (cxds, bcds, hybrid)	Bais, A. S., and D. Kostka. "scds: Computational Annotation of Doublets in Single-Cell RNA Sequencing Data." *Bioinformatics* 36.4 (2020): 1150–1158.	github.com/kostkalab/scds	cxds is the only method without artificial doublets; hybrid is an ensemble
Solo	Bernstein, N. J., et al. "Solo: Doublet Identification in Single-Cell RNA-Seq via Semi-Supervised Deep Learning." *Cell Systems* 11 (2020): 95–101.	github.com/calico/Solo	Neural-network method; top mean AUROC in Xi and Li
DoubletDetection	Gayoso, A., and J. Shor. *DoubletDetection*. Zenodo, 2018.	github.com/JonathanShor/DoubletDetection	Clustering + hypergeometric test; Python
vaeda	Schriever, H., and D. Kostka. "Vaeda Computationally Annotates Doublets in Single-Cell RNA Sequencing Data." *Bioinformatics* 39.1 (2023): btac720.	github.com/kostkalab/vaeda	Variational autoencoder with PU-learning

Tool	Reference	Link	Justification
Chord	Xiong, K.-X., et al. "Chord: An Ensemble Machine Learning Algorithm to Identify Doublets in Single-Cell RNA Sequencing Data." *Communications Biology* 5 (2022): 510.	github.com/13308204545/Chord	Ensemble of existing methods; did not always beat its components in Zhang et al. (12 samples)

Two trivial baselines, library size and number of expressed genes, will be included, as in Xi and Li (2021).

Excluded: the RADO preprint, it is not peer-reviewed yet. Tools needing other data types (OmniDoublet and COMPOSITE for multimodal/multi-omic data, ImageDoubler for images). demuxlet-type genotype methods, which are experimental labeling rather than count-matrix detectors and serve as ground truth. scran's doubletCells is not run separately: its updated version ships as computeDoubletDensity in the scDblFinder package and is run there as a variant (Germain et al.). DoubletDecon gives no doublet scores (so no AUPRC or AUROC), it failed on 4 of the 16 real datasets and most synthetic ones in Xi and Li.

4. Benchmark Datasets

All are public and come from the 16 real datasets with experimentally annotated doublets assembled by Xi and Li (2021), plus simulation. scDblFinder's parameters were optimized on these 16 datasets (Germain et al. 2022), so its results on them may be optimistic. I therefore also simulate doublets from independent singlets and, as a stretch goal, use singlets from DNA-barcoded datasets identified with singletCode (Zhang et al. 2024; GEO accessions listed in that paper).

Dataset (source)	Description	Added variation
Cell hashing: pbmc-ch, cline-ch (GSE108313), mkidney-ch (GSE140262)	Human PBMC (15,272 droplets, 16.7% doublets); 4 human cell lines (7,954; 18.4%); mouse kidney (21,179; 37.3%)	Two species, high doublet rates
Species mixture: hm-12k, hm-6k (10x Genomics hgmm_12k, hgmm_6k)	Human HEK293T + mouse NIH3T3; 12,820 and 6,806 droplets; 5.7% and 2.5% doublets	Heterotypic-only doublets, low rate
demuxlet: GSE96583 (5 PBMC samples), J293t-dm	Lupus patient PBMCs, 3,298–13,916 droplets, 3.4–11.7% doublets; Jurkat + HEK293T, 500 droplets	Disease context, small sample sizes
MULTI-seq: GSE129578 (pdx, HMEC ×2, HEK-HMEC, nuc)	Human/mouse PDX, primary mammary epithelial cells, cell-line nuclei; 5,578–26,426 droplets, 4.6–31.0% doublets	Tumor/xenograft and nuclei samples
Synthetic (scDesign/Splatter; Xi and Li code and Zenodo 4062232)	Known doublets with controlled doublet rate, depth, cell-type number and heterogeneity	Controlled stress tests

5. Evaluation Metrics (committed)

1. AUPRC against experimental doublet labels.
2. AUROC against experimental doublet labels.
3. Precision, recall and TNR at 10/20/40% identification rates, and F1 at each tool's default call.
4. Error in estimated doublet rate (default calls vs. labeled rate).
5. Robustness to misspecified expected doublet rate: tools are rerun with it set to 0.5×, 1× and 2× the true rate, reporting precision, recall, F1, number of cells removed (used also by Zhang et al).

6. Agreement between tools (correlation of found doublets)
7. Wall-clock time.
8. Peak memory.
9. Scaling with droplet number.
10. Usability: installation, documentation, failed runs.

If situation permits, these metrics might be included:

1. Robustness on simulations (doublet rate, depth, cell types, heterogeneity).
2. Stability under 90% droplet/gene subsampling and across random seeds.
3. Downstream DE recovery (DESeq2, MAST, Wilcoxon) after doublet removal.
4. HVG agreement (Jaccard) after doublet removal.
5. Cluster recovery (spurious clusters, singlet purity).
6. Trajectory accuracy (Slingshot/TSCAN) after removal.

Biological validation: experimental ground truth (hashing, demuxlet, MULTI-seq, species), plus checking that predicted PBMC doublets co-express markers of distinct lineages.

Statistics: Friedman test with post-hoc pairwise paired Wilcoxon signed-rank tests (multiplicity-corrected) and Cliff's delta effect sizes.

6. Novelty Statement

Xi and Li (2021) conducted a systematic benchmark of doublet-detection methods, but several newer methods, including Chord and vaeda, have emerged since then. Zhang et al. (2024) extended this work by evaluating a limited set of methods using DNA-barcoded singlets to simulate doublets. This study will compare a broader range of eight methods than Zhang et al, across experimentally labeled datasets from multiple technologies. Also this study will provide neutral, third-party benchmark of the newer tools, alleviating fine-tune issues (scDblFinder’s tune issue, Section 4). In addition to detection accuracy, I will evaluate computational efficiency, usability, and robustness.

7. Task Assignment

I (Cahid) will do all of the tasks.

8. Feasibility Statement

All tools have github pages and clear install instructions. 4 of the 8 tools are written in R. This might be an issue as I didn’t use R before. But with AI assistance, I believe I can run all the tools. The tools that require GPU can be run on TRUBA, but I also have access to a computer with GPU.

9. AI Disclosure

I relied on AI in searching papers and tools, and biological background that I don’t know.

Other references: Xi, N. M., and J. J. Li. *Cell Systems* 12.2 (2021): 176–194, cell.com/cell-systems/fulltext/S2405-4712(20)30459-2. Zhang, Z., et al. *Cell Genomics* 4.7 (2024): 100592. Stoeckius, M., et al. *Genome Biology* 19 (2018): 224. Kang, H. M., et al. *Nature Biotechnology* 36 (2018): 89–94. McGinnis, C. S., et al. *Nature Methods* 16 (2019): 619–626. Zheng, G. X. Y., et al. *Nature Communications* 8 (2017): 14049.